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ARTICLE

Artificial intelligence for web development: Perspectives from the industry

Pyry Pohjalainen[1] | **Juho Vepsäläinen**[2]

[1]Otakaari 1B, Espoo 02150, Finland

[2]Department of Computer Science, Aalto University, Espoo, Finland

Correspondence
Juho Vepsäläinen, Department of Computer Science, Aalto University, Espoo, Finland.
Email: juho.vepsalainen@aalto.fi

Present address
Pyry Pohjalainen,
Juho Vepsäläinen,Department of Computer Science, Aalto University, Espoo, Finland

Abstract

As a field, web development is roughly 30 years old, and during this period, it has been transformed several times already as it has moved from static websites to dynamic web applications. Now, with the introduction of Artificial Intelligence (AI), the field is again at the cusp of a transformation as the latest AI tools might change how to develop for the web yet again. The objective of this study is to look into this phenomenon and understand how AI is changing web development. To achieve this task, we chose to use the sequential qualitative–quantitative design method that combines interviews with a survey to validate and expand our findings from the interviews. We found that AI is used by web developers to increase their development efficiency, as even the current tools are easy to use and access, although they come with several minor downsides, including AI not being able to understand complex logic, the need for validation of AI output, and suggested code that could potentially lead to security issues. While there are clear benefits to using AI tools for web development and AI proficiency is a vital skill for web developers, there are still open questions related to the quality of code produced by AI tools.

INTRODUCTION

As shown by Abrahamsson et al. (2024), the usage of Artificial Intelligence (AI) for web development is a research topic receiving plenty of interest as researchers are figuring out how to integrate ChatGPT and similar solutions into the software engineering workflow. Upadhyaya (2024) suggests that AI could be used to automate repetitive and tedious tasks, while Yilmaz and Karaoglan Yilmaz (2023a) showed that AI can improve the thinking skills and confidence of undergraduate students.

Web technology runs the modern economy

In the early 1990s, Tim Berners-Lee envisioned the World Wide Web (or just web) as a barrier-free medium connecting people globally (Berners-Lee et al. 1994). Since then, the web has transformed the global economy through its different generations: Web 1.0, 2.0, and 3.0. While the Web 1.0 period could be characterized as "read-only," with its static webpages[1] and limited interactivity, it enabled e-commerce to emerge Ariguzo, Mallach, and White (2006). Web 2.0, the democratic Web, allowed users to interact

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with data and the first social media networks, such as Facebook or MySpace, became available (Murugesan 2009). Web 3.0, defined as the semantic web (Murugesan 2009, p. 3-4), was about making the web accessible in nonbrowser applications (Murugesan 2009, p. 16), and specifically, it was about decentralization amongst other, more specific themes (Bharadiya 2023). It is debatable whether Web 4.0 exists, and you can consider that the ideas behind Web 3.0 represent the current state of the art for this article. Most importantly, the web is at the core of the current economy, and it represents a significant driver for business due to its global reach. Earlier web development has been largely manual work, but the emergence of AI tools in recent years has begun to change this.

Why research the usage of AI in web development?

So far, research in the space has focused on what AI could do (Abrahamsson et al. 2024; Upadhyaya 2024) or how good AI is at a specific aspect of web development (Yilmaz and Karaoglan Yilmaz 2023a), leaving a research gap on how developers utilize AI in practice and what the effects of said utilization are. Therefore, focusing on people can give a new point of view for discussing the usefulness of AI at a practical level, which in turn may help to discover new use cases for AI in web development.

Research questions

To understand how web developers are using AI, we have formulated one core research question (CRQ) and four additional research questions (ARQs) that support the core one as follows:

CRQ *How has the utilization of AI affected web development?* This broad question sets the foundation for the research by examining the overall effect the utilization of AI has had on web development, allowing the exploration of various effects, which could be discovered during the interviews or the survey.

ARQ1 *How has AI made web development easier?* This question is important because it directly addresses a key concern for developers—how AI simplifies their work or a part of it. It also prompts an investigation into specific ways AI might affect specific tasks in web development.

ARQ2 *Why are/aren't web developers using AI?* By exploring the usage rates and reasons behind the use or avoidance of AI, this question provides insight into real-world challenges, benefits, and potential problems.

ARQ3 *What is AI good at in the field of web development?* This question identifies the strengths of AI in web development, which could help web developers to find new ways to utilize AI in their craft.

ARQ4 *At what point in the web development process is it most advantageous/disadvantageous to use AI?* This question explores when AI provides the most value during the development process and whether there are stages when AI is not advantageous to use.

Research approach

To research the topic, we chose to use the sequential qualitative–quantitative design method described by Kelle (2006). The method combines qualitative interviews with a quantitative/qualitative survey informed by these, therefore allowing the survey to be tuned while relying less on guesswork on what to ask. This approach allowed us to reduce the amount of bias to gain a clearer understanding of the role of AI in web development.

Structure of the article

First, we focus on the interviews in Interviews section before exploring the survey in Survey section. In Discussion section, we will discuss main themes observed based on this empirical work before concluding in Conclusion section and collecting our answers to the research questions before proposing further research directions.

INTERVIEWS

To calibrate our survey and to gain insight into how practitioners use AI already, we conducted four interviews in December 2024. The interviewees were selected based on their prior experience with using AI for web development. We chose the interviewees from our personal contacts in the field and included developers with varying amounts of experience with using AI for web development. The key point was that the developers had already been exposed to the technology and had formed their initial opinions on its usage. We have listed anonymized descriptions of our four interviewees in Table 1.

TABLE 1 The four interviewees of our study and their technical background.

Interviewee	Degree	Experience (in years)	Role
Interviewee 1	MSc	5	Frontend developer
Interviewee 2	MD	20	Startup founder and developer
Interviewee 3	MSc	8	Fullstack developer
Interviewee 4	BEng	5	Frontend developer

Interview setup

The four interviews were performed online via Zoom in December 2024 in either Finnish or English, depending on the preference of the interviewee. The interviews were conducted as semi-structured interviews, where the interviewer has a set list of questions but can ask elaborating questions from the interviewee to gain further insight as per Doody and Noonan (2013). We used the questions below for this purpose:

1. Have you used AI when working on web development?
2. How often do you use AI?
3. What is your opinion on AI in web development?
4. Give one example of a recent case where you have used AI to try and solve an issue.
5. What kind of problems do you usually solve with AI?
6. What kind of problems do you think AI is good at solving?
7. What kind of problems do you think AI is bad at solving?
8. During what part of the development process are you using AI?
9. Do you think there's an ideal stage in the development process to use AI?
10. Do you think there's a non ideal stage in the development process to use AI?

Coding interview results

To quantify results, we coded the interview transcripts axially using the Atlas.ti software after the completion of the interviews as this allows easier assessment. As the aim of this study was to highlight the effect of the utilization of AI on web development, no hypothesis was made beforehand. This was done to incorporate the interviewees' subjective perspectives into the subject and any personal opinions on the matter. Allowing interviewees to freely express their self-perceptions, without imposing predefined interpretations, enabled the research to identify emerging trends rather than forcing comparisons with existing assumptions. An inductive approach was used alongside grounded theory, as this study is trying to find an unnoticed pattern in how AI utilization has affected web development, and this combination allowed results to emerge organically.

Coding process

For the coding process, we first marked quotations of each answer and then used these quotes to create codes to allow grouping them. In this process, a quote could be included in multiple different codes, and the created codes included findings such as "code quality with AI," "using AI during implementation," and "ChatGPT." After creating codes based on the answers, we created 10 categories. The code categories were made by matching specific codes together to form a group of similar topics, and one code could be included in multiple categories. At the end of the axial coding, there were a total of 171 quotations with 39 specific codes and 10 code categories. These codes ranged from different AI software used in web development to different programming tasks.

Findings

We will go through each of the 10 categories discovered through the coding process next and discuss them one by one.

1. Uses for AI in web development:
The first category consists of the different ways the interviewees described tasks done with AI. The most common way to use AI in web development in the responses was code creation. Other ways mentioned were debugging, data retrieval, formatting, explaining code, rubber ducking, planning code, problem solving, code validation, and writing documentation/comments. How the interviewees described their use of AI in web development covered almost all the typical daily operations of a web developer. The big outlier not mentioned in the interviews was testing code using AI. Interviewee 1 stated that they could write tests using AI but had not done so and would not trust AI for the finalization of code.

2. What is AI good for in web development?:
The second category consists of all the tasks the interviewees described positively during the interviews. The most common task the interviewees stated that AI is good for is simple boilerplate code, which takes time but is easy to code. Other tasks described as specifically good were starting a project, problem solving, batch changes,

planning, and data retrieval. Although the list of tasks deemed suitable for AI use is shorter than the full list of tasks interviewees used AI for, it makes sense to include all tasks under the category of "Uses for AI in web development." This is because developers would not use AI for tasks that do not contribute to development.

3. Issues with AI in web development:
The third category had more quotes compared to the previous category. The previous category had 28 quotations, while the third category had 41 quotations, and it is possible that the interviewees focused more on the negatives than the positives. The most common task in which the interviewees think AI performed poorly was programming complex code that AI cannot understand. Complex code had some differing opinions when comparing the responses between the interviewees. Other issues with AI in web development were up-to-date documentation, best programming practices, newer programming languages, and frameworks.

4. AI in web development phases:
The fourth category compares the number of times an action was described during the interviews, which would fit into one of the specific categories of development phases: planning, design, implementation, testing, deployment, and maintenance. The four stages that could be interpreted from the interviews were design, implementation, testing, and deployment. The phase with the most quotations was implementation, with 21 quotations, the second most was design with 10 quotations, third was testing with five quotations, and finally deployment with two quotations. The discussion around implementation was mostly positive, with minor critiques for the use of AI during implementation coming from AI not understanding complex code and AI using old versions of libraries during project setup.

The most common way in which AI usage was described during design was to create an architectural design and then validate it or request improvements using AI. Negatives for the design phase included the issue of not knowing enough to be able to give AI sufficient details, and without sufficient details, the AI could misdirect the programmer. Testing had a mixed reception, with some positive mentions being that batch changes to Docker files still help work during this phase. Other comments included Interviewee 1 not trusting AI in testing finalization, as AI makes mistakes, and Interviewee 4 mentioning that oversights found during testing are too complex for AI to be of use. Finally, Interviewee 2 mentioned during the interviews that they use AI in almost all of the development phases.

5. How often is AI used?:
The fifth category was the smallest of the code groups, with five different quotations. This category was used to track the answers to Question 2. Interviewees 1 and 2 said they use AI daily in their work, whilst Interviewee 3 said they use AI almost daily, and Interviewee 4 said they use AI a few times a week.

6. AI software:
The sixth category compared the number of times an AI software was mentioned during the interviews. The most mentioned AI software was ChatGPT by OpenAI, with 19 mentions; the second most common was GitHub Copilot with seven mentions. Other softwares in the order of times mentioned were Bloop, Cursor, CLine, and Tldraw. Discussion on AI software by the interviewees came from comparing ChatGPT with other AI software that can access the codebase. Examples of these types of AI software were GitHub Copilot and Bloop.

7. Positive effects of AI usage:
The seventh category kept track of all the positives of using AI in web development. The quotations for this category consisted of three different codes. These codes, in order of prevalence, were as follows: AI makes work faster (12), AI makes work easier (10), and can trust that AI makes simple things correct (9). The code can trust that AI makes simple things correct is the same as for category two, "What is AI good for in web development?" All the interviewees had quotes for all three of these codes.

8. Negative effects of AI usage:
The category includes all the negatives that have arisen from the use of AI in web development. The most common negative effect of AI usage that the interviewees pointed out was mediocre code quality. Other negatives were that AI gives wrong or incorrect information, old formats of code created by the AI, possible security issues from AI code, and the interviewees being skeptical of AI usage in specific scenarios.

9. Effects of AI on web development:
The ninth category focused on how AI has changed working in web development. For the effects, interviewees mentioned programmers slacking, changing work practices due to AI, annoyance caused by AI tools, organizational issues due to AI, and the increased isolation of programmers.

10. Future of web development with AI:
The tenth and final category gathered the perspectives of the interviewees on the future of web development. Only

Interviewees 2 and 4 discussed the future of web development with AI. Interviewee 2 did not believe that web developers would disappear within the next 5 years, even if industry leaders might claim so, although we may have more amateur coders in addition to professional ones. Interviewee 4 remarked that they are still waiting for AI to steal their job and felt that the sky is the limit with the advancements to come, as they could imagine that almost all tasks could be done with AI at some point, but perhaps at the cost of reliability.

Summary

The four interviews gave us insight into how web developers feel about the usage of AI for web development, and we were able to identify several trends. We found that our interviewees used AI in almost all tasks in web development, from planning code to validating code and creating comments. The main takeaway from the interviews was that AI can make simple code faster and with less effort than people, but has some issues with more complex code. The most used AI software was ChatGPT, and the second most used was GitHub Copilot[2]. Also, based on the interviews, the code created by AI should be taken with a grain of salt, as AI can make mistakes, and the results should be validated by a human. The positive effects of AI usage mentioned during the interviews were that the work of the interviewees became easier and more efficient, with the downside of decreased code quality, AI being slow to adopt new changes, such as up-to-date documentation, and other less mentioned negatives. These findings gave us a way to calibrate our survey, which we will discuss in detail within the next section.

SURVEY

Based on our prior interviews, we crafted a survey performed during January and February 2025. We targeted web development professionals in particular to expand on the initial findings of the interviews.

Survey setup

To construct the survey, we created a set of 35 questions available in the Supporting Informaion. The survey was conducted in English, and the Webpropol platform was used to capture the responses. The survey received 114 responses, with 87 of the respondents working in web development and matching the target group.

Analysis procedure for qualitative results

The coding process for the surveys follows a similar model to the process during the interviews, with some changes. The survey data were coded using Atlas.ti software after the completion of the survey. Coding open questions in the survey allowed the finding of patterns in the data that could otherwise have gone unnoticed. No hypothesis was done beforehand, as the survey can be seen as a continuation of the interviews. An inductive approach was used for most questions, as this study is trying to find an unnoticed pattern in how AI utilization has affected web development. Some of the questions utilized sentiment analysis to analyze the general feelings of the respondents about AI, and axial coding was used for analyzing the interviews.

Coding process

The coding process started by going through each open survey question and marking quotations of the responses given by the respondents. These quotes were then used to create codes that allowed us to group similar quotes. In this process, a quote could be included in multiple different codes. The difference between this process and the interview coding process was that the quotes were not shared between the questions, as each question was analyzed with its quotes, codes, and code groups.

Questions 34 and 35 used sentiment analysis to analyze the general feelings respondents have about AI. For these questions, sentiment analysis was used by creating three different codes as follows: positive, neutral, and negative. Each response was grouped into only one of the three categories depending on the emotional tone of the response. This analysis will provide some guidelines on how people working in web development feel about AI.

After all open questions were coded, code categories were created based on the codes created, excluding the three codes used for sentiment analysis. The code categories were made by matching specific codes together to form a group of similar topics. One code could be included in multiple categories. The categorization of the codes helped visualize the 34 main categories of the survey and gave insights into the CRQ.

Analysis procedure for quantitative results

The quantitative data analysis process started with cleaning the data. The goal of the quantitative analysis was to outline patterns and trends in the responses. These trends,

accompanied by qualitative coding, provide an answer to CRQ and our ARQs.

Webpropol platform provides the percentages on responses with alphabetical responses and average, median, and standard deviation for responses with numerical values. Filtering based on answers to quantitative questions allowed the possibility to find hidden patterns, which would not have been findable otherwise. As the survey had 87 responses matching the target group, filtering based on the responses to quantitative responses needed to be done with caution to keep the sample size applicable. Good questions to use filtering were, for example, the age range (Question 2), job titles (Question 3), using AI in web development (Question 4), and how often the respondent uses AI in web development (Question 5).

Findings

A total of 822 people opened and 198 people started responding to the survey. The survey had 114 responses, with 87 of the respondents working in web development, although one of these was filtered out due to inappropriate content, bringing the total amount to 86.

Background of respondents

Almost half of the respondents were between the ages of 20 and 29 (45.4%), with 39 respondents. The age group 30–39 was 27.9% of the respondents, with 24 people. The age group 40–49 was 20.9% (18 respondents), and over 50 years was 5.8% (five respondents). As the age range of people over 50 years old had only five responses, their results should be combined with the 40–49 age group to have enough quantifiable data.

Most of the respondents identified as full-stack developers, with 72.1% of the responses. Other popular options were front-end developer (39.5%) and architect (37.2%). Back-end developer received 29.1% and designer 20.9% of the responses. Respondents were able to select multiple roles, as some roles could not be described with a single option. The total number of respondents to Question 3 was 86, and the total number of selected responses was 171.

Older respondents were also more likely to describe their job role as architect compared to the age group 20–29. The 40 to over 50-year-old group had 60.9% of respondents answer architect (14 responses), while the age group of 20–29 only had 17.9% (seven responses). The exact opposite was true for a front-end developer. The 40 to over 50-year-old group had 26.1% of respondents answer front-end developer (six responses), while the age group of 20–29 only had 41.0% (16 responses). The comparison of job roles for different age groups can be found in Figure 1.

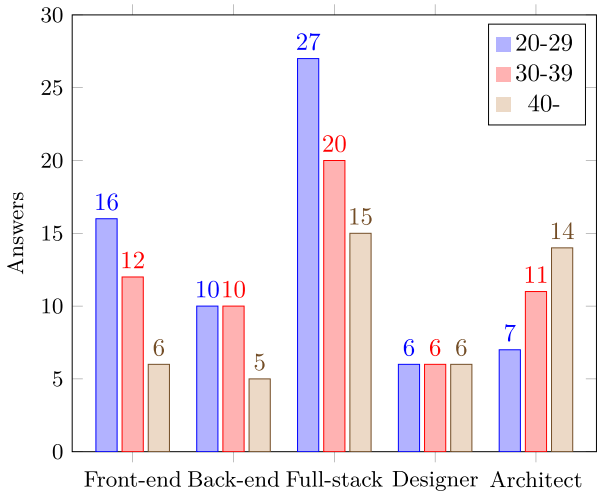

FIGURE 1 Question 3: Which ones of these options would describe you? Older respondents were more likely to describe their job role as architect, and younger respondents were more likely to describe their role as front-end developer. The most common answer was full-stack developer, with 72.1% of respondents.

Approximately, three-quarters of the respondents in the target group have used AI in web development. A total of 65 respondents answered yes to Question 4 (75.6%) whilst only 21 respondents responded with no (24.4%). Age range 20–29 had the highest percentage of not using AI, with 33.3% out of 39 respondents. Age range 30–39 had the second highest percentage of not using AI, with 20.8% out of 24 respondents. The combined age group of 40–49 and over 50 years old had only 13% not using AI out of 23 respondents.

Questions for respondents not using AI

This category included Questions 11 and 12, and respondents were shown these questions if they answered no to Question 4. The responses to Question 11 on why respondents were not using AI in web development could be categorized into three different code groups as follows: company, personal, and ethical issues. For Question 12, the average response from 1 to 10 if the respondent would try AI in the future was 3.6, and the median answer was 3 with a standard deviation of 2.7.

Background on AI usage

This section included Questions 5 through 10. The respondents were shown these questions if they answered yes to Question 4.

Question 5 had a total of 65 responses. The most common response was daily, with 55.4% of the total answers. Weekly had the second most answers with 24.6%. Always had 15.4%, whilst monthly had 4.6% of the total answers.

Question 6 asked respondents if AI has made their work easier. Fifty-seven of the 65 respondents said that AI has made their work easier (87.7%). Eight of the 65 responded that AI has not made their work easier (12.3%). Question 7 continued with the theme from Question 6 and asked respondents how AI has made their work easier and to give one key reason. The responses to this question can be split into the following three categories: AI in programming, AI makes work faster, and AI as a helper.

Questions 8 and 9 focused on the possible increase in efficiency that AI has created. 92.3% of the respondents answered yes to the question "Has AI made your work more efficient" (60/65) and only 7.7% answered no (5/65). Question 9 asked respondents who answered yes to AI making their work more efficient to estimate their efficiency increase from 1 to 10 in increments of 0.1. The average response was 2.9 times more efficient, and the median response was 2.0.

Question 10 asked respondents to tell their three most common use cases for AI in web development. In total, 144 quotes were created from this question, and a total of 27 codes were created. The codes were grouped into six different categories as follows: programming, data, documentation, learning, planning, and testing.

AI tools

The findings in this section focus on answers to Questions 13 through 15. Question 13 asked respondents to choose all the AI tools they have used during web development. The list included all the AI tools mentioned during the interviews and other tools found during research. The most used AI tool was ChatGPT, with 57 out of 64 (89.1%) respondents saying they have used it during web development. The second most common was GitHub Copilot with 45 out of 64 respondents (70.3%). Cursor had 19 (29.7%) and Google Gemini had 15 (23.4%) respondents who have used the tool during web development. Each of the other tools had one response, except Bloop, which did not receive any responses. Due to the timing of the survey in early 2025, we likely missed the impact of Anthropic's Claude, as it seems to have captured roughly one-fifth of the global LLM market at the time of writing (Elad 2025). The results can be found in Figure 2.

Respondents could also answer other AI tools that they have used in web development in Question 14. A total of 24 different AI tools were mentioned, and one respondent mentioned the company's internal AI tools, for a total of 25 different AI tools not mentioned in Question 13. Forty-seven mentions were given in 36 answers. The most common answer was Claude with 19 mentions. V0 had three mentions, whilst Microsoft Copilot and Llama AI had two mentions. All other tools had one mention.

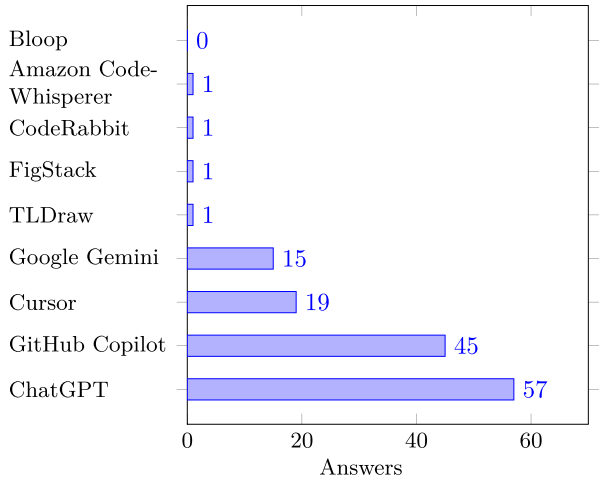

FIGURE 2 Question 13: Which AI tools have you used during web development? The most used tools were ChatGPT and GitHub Copilot.

Question 15 asked the respondents to explain what makes a good AI tool for web development. This question received a total of 49 responses, and 39 codes were created from the responses. Three code groups were created from these codes: output, technical aspects, and usability.

AI processes

Questions 16 through 18 compared how AI performs in web development processes and why. These questions were only shown to respondents who said that they use AI in web development. A total of 65 people responded to Question 16, and Figure 3 shows the responses in detail.

All the processes had a good number of responses. Writing simple[3] code received the most responses, with a response from all respondents, followed by bug fixing and writing complex code, with both receiving 63 responses (96.9%). Design creation/validation received the least responses with 46 responses (70.7%), followed by batch changes with 47 responses (72.3%). Writing simple code also had the highest average score with a 4.5. The second highest was a three-way tie with commenting code, idea creation/validation, and testing, with a 3.6. At the bottom of the results, writing complex code had the lowest score with 2.3, followed by project planning and design creation/validation with 2.6.

Respondents were also able to rate other processes, which were not mentioned on the option list. A total

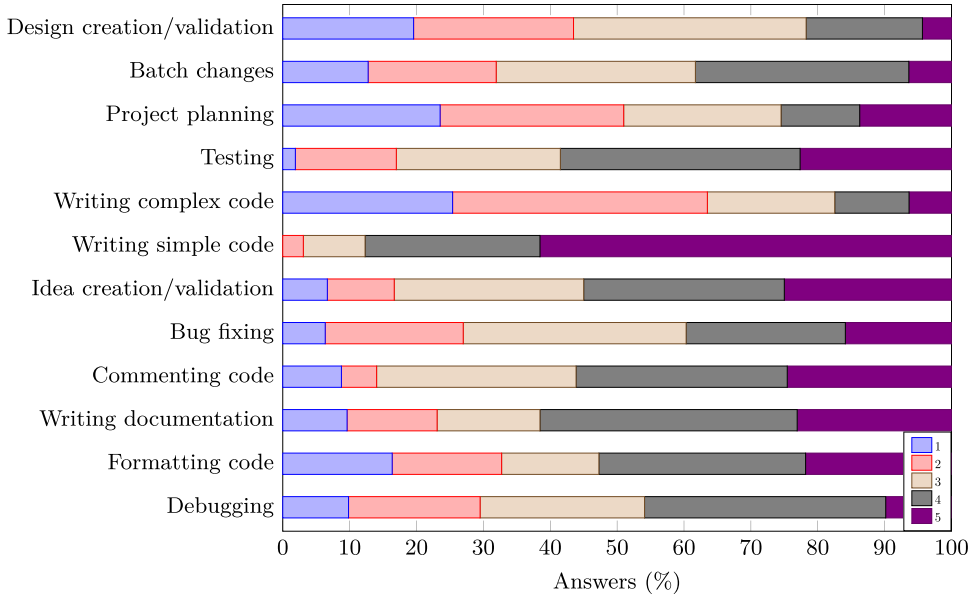

FIGURE 3 Question 16: Grade processes which you have used AI on from 1 to 5. Writing simple code was the most used process and received the highest average rating with a 4.5. Writing complex code had the lowest score with 2.3, followed by project planning and design creation/validation with 2.6. The answers were normalized per question to show their relative amount to each other.

of four responses gave a value to the other process, and three of the responses mentioned a task not on the option list. The three responses given were auditing (5), summarizing information (5), and propagating minor parameter changes through deep function chains (4). One response ranked the other as one without giving a text answer.

Questions 17 and 18 asked respondents to explain why the chosen processes are either good or bad to use with AI. Respondents were asked to give up to three examples of why the categories chosen were either good or bad. Some respondents categorized the responses under each process they rated as good/bad, others gave a more general response, which encompassed all the selected processes, and this caused some problems in categorizing quotes under each category.

Coding all of the responses to Question 17 gave a wide range of responses consisting of 72 quotes in five code categories. These categories were coding, documentation, planning, testing, and technical aspects. Coding category consisted of respondents mentioning how good AI is at creating simple code, and they can create the base code with AI. The coding category was the largest of the categories, which is also explained by the average rating of writing simple code compared to other processes listed in Question 16. The documentation category included codes for commenting on code and writing documentation. Respondents recalled how easy it is to comment/create documentation with AI. The planning category had the fewest quotes of the categories, which is understandable, as project planning had one of the lowest average ratings in Question 16. The testing category can be summarized as AI being good at creating basic unit tests and test data, as no other forms of testing were mentioned in the responses. The technical aspects were the generalized category, in which answers could fit into any of the processes mentioned. The answers included respondents saying that AI is fast, AI knows a lot of technologies, knows the context, gives rapid feedback, and so forth.

Continuing with the coding of Question 18, a total of 55 quotes were created with four different code categories. These categories are coding, issues with AI, planning, and documentation. Coding category consisted of respondents mentioning how AI is poor at creating complex code and how users need to validate the code created by AI. Other responses mentioned that AI uses old syntax and does not follow the existing logic or style of the project. Issues with AI were a generalized category, where answers could fit into any of the processes mentioned. The most commonly mentioned issues were that AI does not understand the problem, has poor output, does not understand context, and the AI hallucinates. The planning category included four quotes, one of which was a response to Question 18. The responses in this category can be summarized as AI being too generic/vague, and users do not trust AI to plan ahead. Documentation was the smallest category, and the category had three quotes. Two quotes from Question 18 were included in this category.

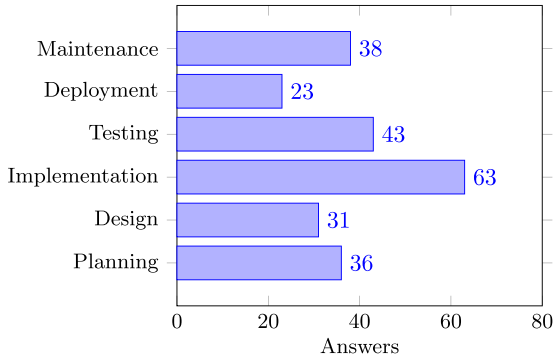

FIGURE 4 Question 19: During which of the parts of the development cycle have you used AI? Implementation was the most used stage for AI, whilst deployment was the least used.

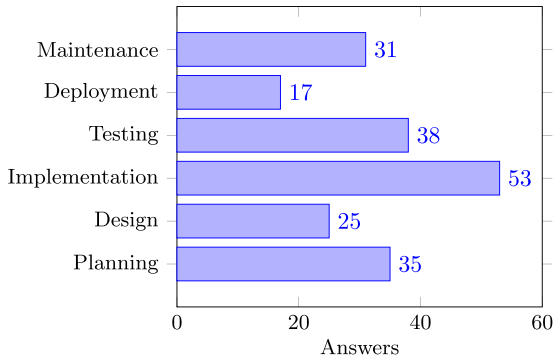

FIGURE 5 Question 20: Which stages of the development cycle are good for using AI? Implementation received the most responses, and deployment received the least amount of responses on which stages are good to use AI for.

Development cycle

Questions 19 through 23 were only shown to respondents who had used AI and these questions were used to answer ARQ4. Question 19 asked when respondents have used AI in the development cycle and the question received 65 responses with 234 options selected. The answers can be seen in Figure 4.

Question 20 asked respondents to rate which phases of the development cycle are good for using AI, and the question received answers from 61 respondents, out of which 199 answers were selected. The answers can be seen in Figure 5.

Question 22 asked respondents to rate which phases of the development cycle are bad for using AI, and the question received answers from 41 respondents, from which 95 answers were selected. The answers can be seen in Figure 6.

Respondents were also asked to explain why the chosen stages in question 20 were good for using AI in Question 21. The same question was also added for Question 22, except

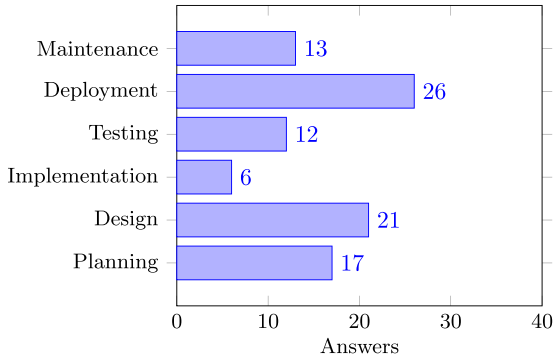

FIGURE 6 Question 22: Which stages of the development cycle are bad for using AI? Deployment received the most responses, and implementation received the least amount of responses on which stages are bad to use AI for.

that it asked the user to explain why the stages are bad for using AI. Question 21 received a total of 24 responses, while Question 23 received a total of 27 responses. For Question 21, five out of the 24 answers mentioned an increase in speed or being faster with the use of AI. Also, all of the stages had speed or being faster mentioned at least once in the results of Question 21 when filtering by each stage from Question 20.

Issues with AI

Questions 24 through 26 asked about AI issues from respondents who have used AI in web development. Question 24 asked respondents how often they receive suggestions from an AI that are incorrect or do not work, and 65 people responded to it. The most selected option was “often” with 40 responses (61.5%). The second most selected option was “sometimes” with 21 responses (32.3%). The least selected option was “rarely” with only four responses (6.2%). From these responses, it can be deduced that AI does frequently give incorrect answers to users.

Question 25 asked if users have received suggestions from an AI that would have caused a security issue if implemented, and 65 people responded to this question. The responses were almost evenly split between yes and no, with no receiving 33 (50.8%) and yes receiving 32 (49.2%) responses.

Question 26 asked respondents to elaborate on their answer to Question 25. The respondents were asked to describe the suggestion that could have caused a security issue if implemented. This question was only shown to respondents who answered yes to Question 25. A total of 22 people responded to Question 26. After coding the responses, a total of four code categories were

created from the codes: code, common security issues, hardcoded, and other. The most common code issues included AI forgetting data validation and authorization. Other code issues mentioned included AI not understanding business logic, AI not following best practices, and incorrect code placement. The common security issues category was the smallest, and the quotes mentioned common security issues of types of attacks that the AI would have caused. The hardcoded category included all the mentions of hardcoded pieces of code, which should not be hardcoded. The responses included mentions of keys, passwords, and environment variables. Other hardcoding-related issues mentioned were that the AI was exposing API keys and not hashing a password when saving to the database. The last category, other, included codes that could not be grouped in the other categories. These codes included AI using less secure encryption, using old or outdated information, and suggesting plugin development.

Trust in AI

Questions 27 through 33 asked respondents to rate each statement from one to five. The questions in this category were shown to all respondents working in web development. The responses to Questions 27 and 28 show that the respondents think that they should be more careful with the data given to vendor-provided AI compared to locally run AI. The answers to Questions 29 through 33 indicate that users should be critical of the data given by an AI. All code written by an AI needs to be supervised by a human, according to the respondents. Also, although the respondents have some trust in AI as a code producer, the respondents trust more in code created by their colleagues rather than code created by AI.

Impacts of AI

Question 34 was only shown to people who have used AI, whilst Question 35 was asked of everyone working in web development. Sentiment analysis was used to analyze the tone of responses and find a general feeling for what people in web development think about AI. For Question 34, which asked how AI has affected your work, the sentiment analysis returned 35 positive, six neutral, and one negative responses. Overall, web developers using AI view it positively as a tool to increase efficiency and help in web development.

Question 35 had a different tone to the responses when compared to Question 34. Question 35 asked what the respondents think the impact of AI in web development is. The sentiment analysis returned a slightly positive verdict, with 24 positive, 20 neutral, and 16 negative responses. Most of the positive responses focused on the current impact AI has had on speed and productivity during the programming process or the lowering of the bar needed to start web development. Some of the positive responses also felt the need to add that although these positive changes are happening, web developers are not disappearing because of this. The neutral responses to Question 35 had a wide range of reasons why the responses were labeled neutral. Some of the main reasons for a response to be labeled neutral were that AI was just another tool, developers having a harder time finding work, and issues with AI. The responses, which were labeled neutral, often mentioned both positive and negative effects of AI, making them neutral in tone. Negative responses to Question 35 focused on AI hype, code quality, and the effect AI will have on working in web development.

Summary

Our survey, calibrated by the earlier interviews, gave us further insight into the usage of AI for web development. Approximately, 75% of the respondents in the target group had used AI in web development. Most of the respondents who use AI use it often, with daily being the most common response, with 55.4% of the responses. Also, 87.7% of the respondents who have used AI said that AI has made their work easier, and 92.3% said that AI has made their work more efficient. The main reasons why respondents did not use AI were that they do not find it useful, it is unethical, or they are not allowed to use it due to security risks.

The best use for AI in web development, according to the responses, was writing simple code, followed by testing, idea creation/validation, and commenting code. The worst use for AI in web development was writing complex code. Respondents also answered that implementation is the most common and useful time, and deployment is the least useful time to use AI in the development cycle.

Overall, people who use AI seem to have a positive sentiment towards AI in web development when the question relates to their work. On the other hand, when all respondents were asked about the effects of AI, the sentiment shifted to neutral.

DISCUSSION

The discussion will first compare the interviews' findings with the survey findings and outline key similarities and differences. Then these findings are compared to past research and the literature. Finally, external and

internal validity for both the interviews and the survey are examined in detail.

Comparing the findings from the interview and the survey

This section compares the findings from our interviews and the survey. The comparison is split into four different categories: AI processes, development cycle, AI tools, and the effect of AI in web development.

AI processes

The task code creation was mentioned the most during the interviews. In the survey, the code creation task was divided into two options as follows: simple code creation and complex code creation. Using AI for simple code creation was the most often chosen option, with every respondent utilizing AI for the task and receiving an average rating of 4.5, the highest rating of any task in the task list. Furthermore, 96.9% of the respondents reported using AI for complex code creation. Despite high usage, creating complex code was rated the least favorable task to use AI for, with an average rating of 2.3. Debugging received the same amount of responses as complex code creation, with 96.9% in the survey. In the interviews, debugging was second in the number of mentions by the interviewees, right after code creation, as mentioned above.

The big outlier between the answers in the survey and the interviews was the lack of mentions for the use of AI in testing during the interviews, as opposed to its many mentions in the survey. In the survey, 81.5% of respondents said they had used AI for testing with an average rating of 3.6, but it was not mentioned during the interviews by any respondent. Batch changes were also discussed in a positive light during the interviews, especially by Interviewee 2, but the results of the survey indicate that batch changes only received average ratings and mentions for AI usage. 72.3% of respondents had used AI for batch changes, and the average rating was 3.0, which was the third lowest rating and second lowest in number of responses.

Development cycle

The implementation phase received most mentions of AI usage in the interviews. This is true for the survey as well: The implementation phase was the most chosen option when discussing AI usage during the development cycle. It had 96.9% usage in the survey, when asked during which stages the respondents had used AI. The three other phases mentioned in the interviews were design, testing, and deployment, in order of the number of mentions. The ordering in the interviews differs from the order in which responses were given in the survey. In the survey, design received the second lowest number of responses with 31, testing received the second most with 43 responses, and deployment received the least number of responses with 23. When comparing how the different stages were described in the interviews, implementation was the clear favorite. The results of the survey support these findings, with 86.9% of respondents answering implementation when asked which stages are good for AI use. The two stages not mentioned during the interviews were maintenance and planning, which were third and fourth in the number of responses for Question 19 ("During which parts of the development cycle have you used AI?").

AI tools

ChatGPT was the most mentioned AI tool in the interviews, with GitHub Copilot receiving the second most mentions. Other tools mentioned during the interviews were Bloop, Cursor, Cline, and makereal.tldraw. ChatGPT and GitHub Copilot received the highest number of responses in Question 13 ("Which AI tools have you used during web development"), which asked respondents to choose all AI tools they had used. ChatGPT received 57 responses, and GitHub Copilot received 45 responses in Question 13. Cursor received the third highest number of responses with 19. Google Gemini was not mentioned during the interviews, but it had the fourth highest number of responses in the survey. Claude was not on the list for Question 13, but it received 19 mentions in Question 14, which asked respondents to write down any as-of-unmentioned AI tools they had used for web development. In terms of the number of responses, Claude is tied with Cursor for the third highest number of responses, but it might have received more responses had it been included in the option list for Question 13. Claude was not mentioned in the interviews.

Effects of AI in web development

As the target group of the interviews was people working in web development, data from the survey on how many of the respondents used AI in web development cannot be compared to the interviews. In the survey, 75.6% of the respondents said that they use AI for web development. People who use AI in web development use it often, with the most chosen option for AI usage being daily, with 55.4% of the responses. Weekly had the second most responses with 24.6% and always had 15.4%.

The interviews found that AI makes work faster and easier for people working in web development. These same results can be seen in the survey, with 87.7% responding that AI has made their work easier and 92.3% of respondents saying AI has made their work efficient. The respondents in the survey said that AI made their work 2.9× more efficient on average. The interviewees mentioned that they can trust AI to make simple things correct, which the results of the survey support. Writing simple code had the highest average rating with a 4.5 out of 5 in Question 16 ("Grade processes which you have used AI on from 1 to 5").

The interviewees stated that humans need to validate the code written by an AI. This statement inspired Question 32 in the survey, which asked respondents to rate the statement "I believe that all code written by an AI needs to be supervised by a human" on a scale from one to five. The average response to this question was 4.4, and the median was 5.0, so it can confidently be said that the respondents in the survey agree with this statement.

Code quality was the most mentioned negative of using AI in the interviews, being mentioned by Interviewees 2, 3, and 4. Code quality was also mentioned in the responses to the surveys in Questions 34 and 35, but the number of responses mentioning code quality was much lower compared to the responses in the interviews. It can be stated that mediocre code quality is a possible issue with the use of AI tools in web development.

Comparing the findings from this study to literature

General benefits of using AI

Upadhyaya (2024) stated that repetitive elements in web development, such as coding, testing, and deployment, can be automated with AI to increase efficiency and minimize the likelihood of mistakes. According to the results of the interviews and the survey, this can be seen to be half true. Respondents to both the survey and interviews said that they generate code using AI, but more complex code is still difficult for AI to generate. This AI-powered code generation increases efficiency according to the responses in the survey.

AI and productivity

In their questionnaire based study, Ngwenyama, Kanita, and Rowe (2025) found that the attitude of developers towards the use of AI changed to be more positive during the 10 weeks the study was performed, while they felt more productive and flourished after the period. The way to interpret this result is that the feeling of well-being and perceived productivity seem to go hand in hand although questions still remain whether there is more to it. The results are in line with our general findings stating that AI makes work faster and easier for web development.

Becker et al. (2025) performed a task-based analysis with experienced developers and open source projects to understand the impact of AI on productivity. Perhaps counterintuitively, they found that although developers believed their productivity had been enhanced, in reality they spent more time on tasks when using AI than they would have without it. This result does not contradict Ngwenyama, Kanita, and Rowe (2025) or our results, but rather it highlights that what developers experience might not reflect reality as the feeling of productivity does not equal measured productivity. It is possible, however, that the framing of the usage, that is, focusing on senior developers with complex tasks, may have an impact on these results and the same goes for tools that seem to improve constantly. The study is a good reminder to research productivity empirically rather than relying only on self-reporting.

AI in software testing

In their research related to software testing, Wang et al. (2024) noted that LLMs are currently only used during the unit and system testing phases. From the results of the survey, the generation of tests was one of the processes in which the respondents thought AI was good at. The responses mentioned that generating tests using AI is fast and easy. Mentions of tests other than unit tests did not appear in our responses, although this could be due to the questions asking for general feedback and not specific feedback on different tests created with AI.

Upadhyaya (2024) stated, "By incorporating artificial intelligence (AI) into testing processes, developers can guarantee enhanced code quality, decrease the amount of time dedicated to manual testing, and enhance the overall dependability of the program." The negatives in the testing stage in the development cycle for the use of AI were test quality and the lack of trust that AI-created test cases are sufficient. In addition, mediocre code quality was seen as a negative in the interviews and was mentioned by some of the respondents in the survey.

AI in learning software engineering

Yilmaz and Karaoglan Yilmaz (2023a, 2023b) highlighted in their research that AI can improve learning among

programming students. Although the context of undergraduate students learning programming and web development professionals learning might be different, and in our opinion, the point still holds. Multiple respondents in the survey mentioned that learning new technologies and working with unfamiliar languages is made easier with AI. Some respondents were wondering if AI would decrease learning, but these responses were in the minority when compared to boosting learning.

MacNeil et al. (2023) used GPT-3 and Codex to generate code explanations for students in an online university course on web development. The study found that the use of AI-generated code explanations helped to explain the code to students (MacNeil et al. 2023). Code explanation was mentioned in the responses to both the interviews and the survey as a use for AI in web development.

Limitations

Interview reliability and validity

Internal validity

The interviews were held in English or Finnish, depending on the preference of the interviewee. Some of the questions did not translate perfectly from English to Finnish, which could cause the interviewee to understand a question differently depending on the language used in the interview.

External validity

Only four interviews were conducted in total. The study could have benefited from a greater number of interviews to get a wider audience for the initial questions, as trying to conclude from the responses of four individual web developers compared to the overall number of web developers is difficult. To gain a more concrete understanding of what the effect of AI has been on web development, the number of interviews would need to be increased, and a diverse cast of interviewees found.

Survey reliability and validity

Internal validity

In the survey, the list of AI tools included AI tools that were mentioned during the interviews and found during internal research. In total, 25 additional AI tools were mentioned in the survey responses on top of the AI tools already mentioned in the options list. The results would most likely be affected if all AI tools mentioned additionally in the survey had been added to the options list to begin with.

When comparing the responses between stages in the development cycle that are good and those that are bad to use AI for, the question to rate stages which are good received approximately 1.5× more respondents than to bad stages. We believe the respondents were reluctant to answer two similar questions, as they may have felt that if they left a particular phase out of the ones listed as suitable for AI use, it was a sufficient action to signal that it is unsuitable for AI use.

As AI is progressing rapidly and new competitors are being published often, some AI tools that could be in use currently might not be found in the results, as the tool might not have been published yet when the survey was open. An example of this would be DeepSeek R1, which was published in the middle of collecting survey answers when most of the respondents had already responded to the survey (AI 2025).

External validity

The survey received 86 valid responses from web development professionals. The survey had a wider reach than the interviews, but once again, trying to compare the opinions of 86 people to the whole group of web developers can be problematic. To gain a more concrete understanding of the effect of AI has had on web development, the number of responses for the survey would have to be increased.

People who are not interested in AI might not be as excited to answer a survey on the effects of AI on web development. This might affect the data for Question 4 ("Do you use AI in web development?"). Also, since respondents were found from the developer and AI communities on Reddit, respondents might be more engaged with these communities than regular web developers. This could affect the results and responses of the survey.

There may be a geographical bias for the respondents in the survey. As no data on the locations of the respondents to the survey was collected, this cannot be confirmed. We believe that there is a chance for a geographical bias as the survey was shared on LinkedIn, Reddit, and Telegram. The respondents who found the survey post on LinkedIn or Telegram are most likely from Finland. This could create a geographical bias as only a small minority of people working in web development live in Finland.

CONCLUSION

The goal of this study was to research the effect that the utilization of AI has had on web development. Next, we will go through our findings per research question before considering potential future research directions.

CRQ: How has the utilization of AI affected web development?

The utilization of AI in web development has affected web development in multiple ways. First, it has increased the efficiency of people working in web development, and it has made web development easier. Both the interviewees and the survey respondents mentioned that AI has started to replace Google when they need to find information on something. Respondents also mentioned that it is easier to create code in languages and frameworks that they are unfamiliar with, as AI can create a good base and give suggestions, which guide the user.

Code quality was of concern among the respondents of both the interviews and the survey. The respondents agreed that AI will cause code quality to decrease as more code can be created faster. The respondents also mentioned the need to keep up to date with their knowledge of AIs as they progress rapidly, which consumes time.

Multiple respondents mentioned the decrease in the number of job openings for junior-level programmers in their responses, and some of the interviewees also discussed AI in terms of employment opportunities. There is a divide between the respondents on the impact that AI will have on jobs in web development. Some of the respondents said that AI will decrease the number of jobs in the future, while others said that it will only affect junior roles.

ARQ1: How has AI made web development easier?

AI making work easier was one of the main findings of the interviews and in the survey, with 87.7% of respondents who used AI in web development saying that AI has made their work easier. In the interviews, the interviewees stated that AI makes their work faster and easier and can help in multiple tasks, such as troubleshooting and bug fixing. Looking at the responses to Question 7 in the survey, 49 respondents answered how AI has made their work easier. The answers to this question are that AI makes work faster, AI helps with programming tasks, and it offers a tool for research and writing documentation. It can be said that AI makes the development process easier.

ARQ2: Why are/aren't web developers using AI?

ARQ2 has two answers. One relates to why developers use AI and others do not. First, web developers are using AI as it makes the web development process easier and more efficient. This can be concluded from the answers in the survey to Questions 6–9. Of the people who responded to the survey, 75.6% of the respondents said that they use AI in web development.

There are three main reasons why some of the respondents did not use AI in web development. These can be categorized as company, personal, and ethical issues. For company issues, multiple respondents stated that they are not allowed to use AI due to restrictions at their workplace and cited security reasons for this. The personal reasons had respondents mentioning that they do not see the benefits of using AI. Other reasons in this category were that the respondents did not trust AI, the cost was too high, some privacy concerns, and that the respondents enjoyed writing code themselves. Finally, ethical issues included environmental issues with AI, regulatory issues, and copyrighted material, which the AI models use.

ARQ3: What is AI good at in the field of web development?

Multiple different processes are good to use AI for in web development. The clear answer is writing simple code[4], which had an average rating of 4.5 out of five in Question 16 of the survey. Looking at the processes in Question 16, we think it is easier to single out the processes that had an average rating of less than three. These processes are writing complex[5] code, project planning, and design creation/validation. The other processes we would say are at least neutral or better. In addition, all processes had at least two responses of five and at least 10 ratings of four or better, so it can also be said that it depends on the person. Some people might find AI more useful for different processes compared to others.

ARQ4: At what point is it most advantageous/disadvantageous to use AI in the web development process?

The most advantageous stage in the development process to use AI is during implementation. This stage was the clear favorite of the respondents to the survey, and the interviews support this claim. The main reasons for the implementation stage were the increase in the productivity of writing code with AI and support for manual tasks. The most disadvantageous stage to use AI was deployment. The respondents mentioned that they would not trust AI

deployment and could not see the value of integrating AI into this stage.

Future research directions

To further investigate the effect the utilization of AI has had on web development, we suggest comparing two web development teams, where one has access to AI to aid in development, while the other does not. We have taken inspiration from Yilmaz and Karaoglan Yilmaz (2023b), as they conducted similar research, except with students and studied the effect of AI on learning programming for undergraduate students. In our opinion, this could outline the differences between a team that can use AI and a team that is not able to.

Code quality with AI in web development could also be an emphasis of study for the future. This research could be conducted as a comparison between code and functions written by a human and AI tools. The human and AI are given the same task, and the results are evaluated by both AI and humans. This research could elaborate more on the code quality differences between humans and AI, as this issue was a main talking point during the interviews, but mentions on code quality decreased in the survey responses.

Given technological change tends to come with reluctance for adoption, it would be worthwhile to understand reasons why developers might avoid using AI for web development. Furthermore, understanding in detail the gaps in usage and the reasons for this would be worthwhile in shaping further research and tool development agenda. Same goes for understanding code quality as generated code may come with issues of its own even if it is faster to produce initially as code correctness is a longstanding issue in software development.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work, we used OpenAI's ChatGPT (GPT4-Turbo) and Grammarly service to suggest improvements to written text and give ideas on titles and subtitles. After using these tools, we reviewed and edited the content as needed, and take full responsibility for the content of the publication.

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest.

ENDNOTES

[1] Early webpages were defined using HTML (HyperText Markup Language), technology still in use.

[2] Given the interviews were held in December 2024, it is likely that, for example, Claude would show up as a tool at the time of writing, as it has become a popular option since then.

[3] Note that in this case, what was considered simple or complex was based on the subjective view of our respondents.

[4] What is considered simple was simple from the respondents' point of view, meaning likely there were differences in the interpretation of this word. It would be a good idea to use more accurate wording for any subsequent studies.

[5] Again, what is complex tends to be subjective and it is recommended to use more specific wording for subsequent studies.

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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AUTHOR BIOGRAPHIES

Pyry Pohjalainen received his MSc degree in computer science from Aalto University, Finland, in 2025. Pyry is a web development specialist and works as a Software Engineer in Helsinki, creating and maintaining microservices used by global industrial companies. Pyry continuously grows expertise in JavaScript, TypeScript, Python, and C#, incorporating AI technologies when possible.

Juho Vepsäläinen received his DSc degree in computer science from Aalto University, Finland, in 2025. Juho's research interests include web development, web performance, and green computing. He contributed to open-source projects, such as Blender and webpack as a core team member, and publishes technical literature under the brand SurviveJS while running the Future Frontend conference series.

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